Viva Presentation — Speaker Notes & Q&A

Early Warning Systems for Banking Crises

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Presentation Overview

13 slides · 10-minute speech · 5-minute Q&A

Structure: Title (0:30) → Problem (1:00) → Design (1:00) → Stage 1 results (0:45) → Prospective collapse (1:00) → Shape functions (1:30) → LOCO/Stage 2 (1:00) → SHAP/Calibration (0:45) → 2007 ranking (0:45) → Hypotheses (1:00) → Implications/Close (0:45)

SLIDE 1 — Title [00:00–00:30]

Open without preamble. Look up, not down. Speak directly to the examiner.

Script:

"This dissertation asks a precise question: why do machine learning models for banking crisis prediction fail under genuinely prospective conditions — not whether they fail, which we already know, but through exactly what mechanism. I will show you the answer using a model class that makes the failure directly visible."

⏱ 30 seconds. Move to Slide 2 immediately. No filler.

SLIDE 2 — The Problem: A Known Failure, Unknown Mechanism [00:30–01:30]

Reference the chart at the bottom. The visual carries the argument — do not read the bullet points.

Script:

"The EWS literature established a clear empirical fact. ML outperforms logistic regression under standard expanding-window validation — but falls below the 0.75 AUROC benchmark under strict temporal testing. The chart at the bottom shows this starkly: the blue bars are what models achieve when tested the usual way; the red bars are what happens when you train exclusively on pre-crisis data. Both RF and EBM fall dramatically."

"But the literature stopped at documenting the numbers. It did not explain the mechanism, test whether interpretable models are more or less vulnerable, or cleanly separate algorithm effects from sentiment. Those three gaps are exactly what this dissertation fills."

⏱ 60 seconds. Point to the chart. Summarise; do not read bullets.

SLIDE 3 — Data, Design and Model Sequence [01:30–02:30]

This is the justification slide. The key point is why EBM is in the sequence — not just what it is.

Script:

"The dataset is the JST Macrohistory Database — 18 advanced economies, 1985 to 2020. After lag construction, Stage 1 covers 569 country-year observations with 37 crisis events at a 6.5 percent base rate. Stage 2, where FinBERT speech sentiment is available, covers 2003 to 2020 with 26 crisis events."

"The model sequence is the core design decision. M1 and M2 are standard benchmarks. M3 adds FinBERT sentiment to XGBoost. The novel addition is M2.5 — the Explainable Boosting Machine. It uses the same 41 macro features as M2 but a fundamentally different algorithm — one that learns exact shape functions rather than trees. This lets me isolate the algorithm effect from the sentiment effect. And crucially, it makes the failure mechanism directly visible. No other model in this sequence can do that."

⏱ 60 seconds. Point to the EBM card. This slide justifies your design.

SLIDE 4 — Stage 1 Results: EBM Leads Across All Metrics [02:30–03:15]

Show the result but plant the caveat immediately. The examiner is waiting for you to acknowledge the inflation risk.

Script:

"Stage 1 results on the extended 1988–2020 panel. EBM achieves AUROC 0.753 — the only model to meet the 0.75 benchmark — with AUPRC 0.273, highest of any Stage 1 model, using identical features and cross-validation design as Random Forest. The chart makes the gap visible."

"But I want to be careful here. These are expanding-window figures. All three models accumulate GFC-period data in later training folds, which inflates all metrics. EBM appears to have solved the trade-off — higher accuracy AND interpretability. The next slide shows what happens when I remove that advantage."

⏱ 45 seconds. Do not over-claim. The caveat is as important as the number.

SLIDE 5 — The Prospective Collapse [03:15–04:15]

This is your most important empirical result. Pause on 0.333. Let it land.

Script:

"This is the central empirical finding. I apply strict temporal validation — Split B: train on 1988 to 2002, test on 2003 to 2011. No GFC data in training. Six positive training cases."

"Random Forest achieves AUROC 0.481 — at chance level. EBM, the model that led every expanding-window metric, collapses to AUROC 0.333. Substantially below random. Worse than Random Forest."

[Pause. Look at the examiner. Let the number sit.]

"The model with the highest in-sample performance and the highest interpretability fails most severely when the regime changes. The bar chart puts this in context: the useful prediction benchmark is 0.75. Random is 0.5. Both models fall far short."

"I want to be honest about a limitation. Split B has only six pre-GFC positive training cases. The result is diagnostic, not conclusive. But combined with the shape function evidence on the next slide — which is entirely independent of Split B — the structural explanation is compelling."

⏱ 60 seconds. Pause after '0.333'. Do not rush through the limitation.

SLIDE 6 — Shape Function Regime Instability [04:15–05:45]

This is the pivot of the entire presentation. Slow down. Point to the figure.

Script:

"This slide contains the dissertation's most theoretically significant finding. I trained EBM twice — on a pre-GFC window 2003 to 2006, and on a GFC-inclusive window 2003 to 2010 — and plotted the shape functions for the same features side by side. That is Figure 4.7."

"Look at term_spread_lag3 — native importance 1.717, the single highest-ranked EBM feature. In the pre-GFC window, the model learns a monotonically increasing risk relationship: higher spreads signal higher crisis risk. In the GFC-inclusive window, that relationship reverses at high spread values. The curves cross."

"The same feature value is associated with reduced risk in one regime and increased risk in the other."

"Now — and this is crucial — because it is EBM, there is no SHAP approximation. If I had done this with a Random Forest, I could not tell whether the difference reflects a genuine model change or an approximation artefact. With EBM, the shape function IS the model parameter. This instability is therefore direct evidence about the data-generating process, not about the attribution tool."

"And here is the mechanical link to AUROC 0.333: the model trained on pre-GFC data applies the blue-line mappings to GFC observations. It gets the sign wrong on the top feature. Every high-spread GFC observation gets assigned low risk. Systematic misclassification follows. AUROC below random is the necessary arithmetic result."

"The panel logistic provides independent confirmation — sign reversal at the same lag horizons, from a completely different analytical framework."

⏱ 90 seconds. This is the pivot. Slow down here. Point to the crossing curves on Figure 4.7.

SLIDE 7 — LOCO and Stage 2 Performance [05:45–06:45]

Script:

"Two results on this slide. On the left — the LOCO validation: hold each country out in turn, train on the remaining 17, test on the held-out country's crisis events. EBM achieves mean AUROC 0.893 against RF's 0.839, with lower standard deviation — 0.215 versus 0.254 — suggesting more consistent cross-country ranking."

"Japan is the notable outlier: both models are near-random, at 0.138 and 0.069. Asset-deflation crisis dynamics are poorly represented in a cross-country training distribution dominated by credit-cycle crises. This is consistent with the crisis heterogeneity argument."

"On the right — Stage 2 AUPRC. The critical comparison: EBM macro-only achieves 0.532. M3 with FinBERT sentiment achieves 0.411. The interpretable macro-only model outperforms the sentiment-augmented one. The algorithm effect is at least as large as the sentiment signal — if not larger."

⏱ 60 seconds. Emphasise the Stage 2 comparison — EBM macro beats M3+sentiment.

SLIDE 8 — Feature Attribution and Calibration [06:45–07:30]

Script:

"Two results on this slide. The pie chart shows SHAP attribution for M3 by lag group. Macro features account for 73.8 percent of total attribution. Sentiment contributes 26.2 percent within the GFC episode."

"On the right — EBM's native feature importance. term_spread_lag3 leads by a substantial margin at 1.717. The top five features are all macro. No sentiment feature appears in the top 20."

"The key finding is convergence: two completely independent methods — SHAP on M3 and EBM native importance on M2.5 — both identify term_spread_lag3 as dominant. Methodological triangulation. But this agreement reflects shared sample-specific correlation within the 2007 GFC episode, not causal invariance across regimes."

"On calibration: EBM achieves the best ECE — 0.084 — without post-hoc isotonic calibration. The ensemble achieves the best Brier Score at 0.056."

⏱ 45 seconds. Keep moving — these are supporting results, not the centrepiece.

SLIDE 9 — 2007 Cross-Sectional Ranking [07:30–08:15]

Script:

"This is explicitly exploratory and post-hoc — stated clearly in Section 4.10.3. But it reveals something important about the model's scope."

"In the 2007 cross-section, countries are classified by crisis type. The mean E1-EBM probability for fundamental credit-boom crises is 0.695. For contagion-driven cases: 0.395. For no crisis: 0.325."

"The ordering makes intuitive sense. A model trained on domestic macro precursors correctly assigns higher risk where those precursors are elevated. It underperforms on contagion-driven cases because cross-border transmission channels are absent from the feature set. This identifies exactly what the model would need to improve: bilateral bank exposure data following Cetorelli and Goldberg."

⏱ 45 seconds. Frame this as scope characterisation, not a new claim.

SLIDE 10 — Hypothesis Verdicts and Four Conclusions [08:15–09:15]

Script:

"Three hypotheses, four conclusions."

"H1: directionally supported. M3 AUPRC 0.411 exceeds M2-matched 0.343. But the CI spans zero, and EBM macro-only at 0.532 outperforms M3. Algorithm choice dominates."

"H2: not supported in strict formulation. No sentiment feature in the top 20 EBM native importance."

"H3: supported and refined. The trade-off is not binary. EBM leads in-sample but fails most severely prospectively. The representational flexibility that enables shape function interpretability is the same mechanism that produces regime-specific overfitting. This is not a frontier you navigate — it is a unified structural explanation."

"Four conclusions together form a single argument: ML-based EWS failure is not a function of model choice, feature design, or data quantity. It is structural instability in the mapping between predictors and crisis outcomes across regimes."

⏱ 60 seconds. Land the four conclusions quickly — they are set up by everything before.

SLIDE 11 — Implications and Close [09:15–10:00]

Script:

"Four practical implications. One: prospective validation must be mandatory — a model risk governance requirement, not an optional academic exercise. Two: EBM should be adopted for its diagnostic value — shape functions tell a regulator when the model's learned mapping has shifted. Three: ensemble averaging is most defensible for calibration — Brier Score 0.056 — not discrimination. Four: sentiment augmentation needs multi-episode validation before any operational deployment."

"Close with this. Memorise it. Do not read it:"

"The central limitation of EWS is the absence of a stable mapping between predictors and crisis outcomes across regimes. ML makes this visible; it cannot resolve it."

"Thank you."

⏱ 45 seconds. Deliver the closing sentence without looking at the slide. Pause after it.

Q&A — 5-Minute Response Window

Six challenging questions. Each answer is calibrated to the time available. Be concise. Acknowledge limitations — do not defend against them.

Q1: Split B has only 6 training positives. How can you draw meaningful conclusions from it?

Acknowledge it directly — this is stated explicitly in Section 4.5.1. Split B with 6 training positives cannot deliver statistically reliable AUROC estimates. The result is diagnostic and exploratory, not confirmatory. However, the shape function analysis in Section 4.7.3 is entirely independent of Split B — it is estimated from the full pre-GFC and GFC-inclusive panels. The two findings — the AUROC collapse and the shape function sign reversal — converge on the same structural explanation from completely different analytical angles. I am not claiming Split B proves regime-specific overfitting. I am claiming it is consistent with the regime instability the shape functions demonstrate directly. ⏱️ 50 seconds.

Q2: EBM AUROC 0.333 could reflect sample thinness rather than regime shift. How do you distinguish between these two explanations?

You cannot fully distinguish them from a single split — this is acknowledged in the dissertation. However, three pieces of converging evidence point toward regime shift rather than pure sample thinness. First: RF also has 6 training positives in Split B yet achieves 0.481 — above chance — while EBM falls to 0.333. If sample thinness were the sole driver, both models should fail similarly. The differential collapse is consistent with EBM's higher representational flexibility encoding the pre-GFC crisis signature more aggressively. Second: the shape function analysis is independent of Split B and shows genuine non-stationarity. Third: Split A shows EBM meeting the 0.75 benchmark, which would be unlikely under pure sample thinness alone. ⏱️ 55 seconds.

Q3: You cannot separate the sentiment signal from the algorithm effect in Stage 2. Does H1 have any validity?

This is the dissertation's own criticism of H1, stated explicitly in Section 4.2 and the hypothesis assessment table. The M3 versus M2-matched comparison confounds algorithm, regularisation, and feature set simultaneously. This is precisely why EBM was introduced as M2.5 — holding features constant at 41 macro variables while varying only the algorithm. The result: EBM macro-only AUPRC 0.532 versus M3 macro-plus-sentiment 0.411. The interpretable macro-only model outperforms the sentiment-augmented one. H1 is therefore characterised as directional evidence only — explicitly not confirmatory. ⏱️ 45 seconds.

Q4: Stage 2 AUROC figures of 0.93–0.96 inflate due to GFC concentration. Are they informative at all?

Yes — but only as in-crisis discrimination measures within the 2007 GFC cross-section, which is stated from Section 4.1 onward and repeated at every result table. Stage 2 answers one specific question: how well does each model rank 18 countries by crisis probability in 2007? That is a meaningful question, even if it is not a generalisation claim. The dissertation explicitly labels all Stage 2 metrics as upper bounds on prospective performance. The more important numbers are the strict temporal validation figures and the Stage 1 multi-episode AUROC where EBM achieves 0.753 on 37 events across 1988–2020. ⏱️ 45 seconds.

Q5: Why not use a regime-switching model directly to address non-stationarity, rather than documenting that a static model fails?

Appendix B answers this directly. A proof-of-concept regime-conditional EBM prototype was implemented in NB11. EBM-R1 on pre-GFC data achieves AUROC 0.410 — below random — because there are too few crisis positives per regime to estimate stable parameters. EBM-R2 in the GFC regime has only 2 training positives — completely unreliable. The statistical constraint binds immediately. The dissertation's contribution is not to solve non-stationarity — it is to characterise the failure mechanism precisely enough that future solutions can be designed with correct targets. Regime-conditional modelling is identified as the most important future extension, conditional on richer crisis episode data across multiple structurally distinct regimes. ⏱️ 55 seconds.

Q6: SHAP already shows feature importance variation across time. What does EBM add that SHAP cannot provide?

SHAP on Random Forest or XGBoost produces approximate attributions. When the evaluation distribution shifts across training windows, SHAP attributions can change for two distinct reasons: the model genuinely learned a different relationship, or the approximation behaved differently under the new distribution. These two explanations are indistinguishable in SHAP output. EBM shape functions have no approximation step — they are the model's exact fitted parameters. Any change across training windows reflects a genuine change in the model's learned mapping, full stop. This is what allows the shape function evidence to constitute direct evidence about the data-generating process rather than about the attribution tool — and it is the entire basis for the claim in Section 6.2.3 that EBM makes the failure mechanism visible rather than merely documenting performance collapse. ⏱️ 55 seconds.

Total Q&A budget: ~5 minutes · 6 questions · average 50 seconds each · leave 10 seconds buffer between answers.